Classifying Ulcerative Colitis Severity Using GANs and Deep Learning

Enhancing Colonoscopy Image Quality and Improving Classification Accuracy with Histogram Equalization and Convolutional Neural Networks

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Abstract—Medical imaging plays a critical role in healthcare, particularly in diagnosing and treating diseases. However, lowquality images with poor contrast present challenges for accurate classification. This project explores the use of Histogram Equalization (HE) and Generative Adversarial Networks (GANs) to enhance contrast in colonoscopy images from the LIMUC dataset. Convolutional Neural Networks (CNNs) and other neural networks are then employed to classify ulcerative colitis severity. By addressing low contrast and class imbalance issues, this study aims to improve diagnostic accuracy and reliability, ultimately benefiting clinical decision-making.

I. Introduction and Statement of the Problem

Medical imaging is a cornerstone of modern healthcare, enabling the visualization of internal structures and aiding in the diagnosis and treatment of various diseases. Among the many applications of medical imaging, colonoscopy plays a pivotal role in diagnosing and assessing gastrointestinal conditions such as ulcerative colitis (UC). Ulcerative colitis is a chronic inflammatory bowel disease characterized by inflammation of the colon and rectum, often requiring regular monitoring and evaluation to determine disease severity. One commonly used metric for evaluating UC severity is the Mayo Endoscopic Score (MES), which classifies the disease into four categories: Mayo 0 (normal), Mayo 1 (mild inflammation), Mayo 2 (moderate inflammation), and Mayo 3 (severe inflammation). Despite its importance, the quality of medical images, particularly colonoscopy images, often suffers due to factors such as low resolution, poor contrast, and variability in imaging conditions. These issues make it difficult to extract critical information, leading to challenges in accurately diagnosing and classifying diseases. Furthermore, the LIMUC dataset, which comprises 11,276 colonoscopy images labeled according to MES, exhibits significant class imbalance, with a disproportionately high number of images in the Mayo 0 category compared to the other categories. The accurate classification of ulcerative colitis severity is crucial for effective patient management, including treatment planning and monitoring disease progression. However, the limitations in image quality and the inherent class imbalance in datasets like LIMUC hinder the performance of traditional machine learning and deep learning models. While methods such as Histogram Equalization (HE) can enhance image contrast, they often introduce noise and may result in the loss of critical medical information. On the other hand, advanced techniques like Generative Adversarial Networks (GANs) provide a sophisticated, data-driven approach to image enhancement by learning from the dataset to generate realistic, high-quality images. Motivated by the need to address these challenges, this project aims to develop a comprehensive pipeline for improving the quality of colonoscopy images and accurately classifying ulcerative colitis severity. By combining image preprocessing techniques (HE and GANs) with deep learning models such as Convolutional Neural Networks (CNNs), this study seeks to enhance both image quality and classification accuracy. Additionally, the project will explore strategies to mitigate the impact of class imbalance, ensuring fair and robust performance across all MES categories. The primary challenges addressed in this project are the low contrast in colonoscopy images, which limits the ability of machine learning models to extract meaningful features and reduces classification accuracy, and the class imbalance in the LIMUC dataset, where the uneven distribution of images across MES categories introduces bias in model training, leading to suboptimal performance in underrepresented classes. To tackle these issues, this project proposes approaches such as enhancing image contrast using HE and GANs to improve feature extraction and image quality, training CNNs and other neural networks on enhanced images to classify ulcerative colitis severity into four categories based on MES, comparing the performance of HE and GANs in preprocessing to identify the most effective approach for improving classification accuracy, and implementing techniques such as data augmentation, class weighting, and focal loss to address class imbalance and ensure balanced learning. By addressing these problems, the project aims to contribute to the development of reliable, accurate diagnostic tools for ulcerative colitis

severity assessment, ultimately supporting better patient outcomes and advancing the field of medical imaging.

II. Review of Literature

The literature on ulcerative colitis (UC) grading and endoscopic image classification presents various innovative methodologies, each contributing to the development of more accurate diagnostic tools. Polat et al. (2023) propose a regression-based deep learning model to estimate UC severity based on the Mayo Endoscopic Score, utilizing computeraided estimation to improve clinical accuracy. Their approach focuses on leveraging deep learning to assess endoscopic images, achieving promising results. However, the model's performance is heavily dependent on the dataset used, and its generalizability to different clinical settings and patient populations is a potential limitation [1]. This work is particularly relevant to our project as it provides a foundation for using regression-based techniques to assess disease severity, which could be enhanced by incorporating additional image features and patient-specific factors. Pyatha et al. (2023) explore the application of Vision Transformers (ViT) for self-supervised learning in UC grading during colonoscopy procedures. ViTs are known for their ability to model long-range dependency in images, which helps in accurately grading UC severity. The paper discusses how self-supervised learning can be employed to train the model on a large number of unlabeled images, reducing the need for extensive labeled datasets. While the use of ViT brings substantial improvements in feature extraction, a major limitation of this approach is its computational complexity, which may hinder real-time implementation, especially in clinical environments with limited resources [2]. This research is useful for our project by introducing advanced transformer models, which could improve the performance of UC severity classification, but we would need to consider optimizing computational efficiency. Lopez et al. (2024) introduce a multi-modal detection transformer combined with data engineering techniques to stratify UC patients. Their model integrates various data sources (e.g., clinical, demographic, and endoscopic image data), utilizing a transformer architecture for patient stratification. This multimodal approach enables better understanding and prediction of UC progression. However, a limitation is the need for careful data preprocessing and normalization to ensure that disparate data sources are effectively integrated. This paper is relevant to our project as it highlights the potential of multimodal learning, which could be useful in incorporating both endoscopic images and patient history for a more holistic prediction of UC severity [3]. Inci (2022) investigates the use of active learning in conjunction with synthetic sample selection to improve endoscopic image classification. The active learning framework identifies the most informative samples for training, while generative models create synthetic data to augment the dataset. This approach helps improve classification accuracy in situations where labeled data is scarce. However, one limitation is that the performance of the model depends on the quality of the synthetic samples generated, and there is a risk of overfitting if the synthetic data closely resembles the original dataset without introducing sufficient diversity [4]. This method can be beneficial for our project by helping address the challenge of limited labeled data, particularly for rare UC cases. C¸aglar et al. (2023)˘ combine active learning with generative data augmentation to enhance the Mayo Endoscopic Scoring Classification of UC. By generating synthetic data to augment the training set, the authors address the problem of class imbalance in UC datasets, which often results in biased model performance. The model also employs active learning to focus on the most informative samples. However, a potential limitation of their approach is that the generative model may introduce synthetic samples that do not accurately represent real clinical scenarios, potentially leading to model inaccuracies [5]. This paper provides valuable insights for our project, particularly for augmenting training data and handling class imbalance, which is crucial in medical image classification. Lv et al. (2023) conduct a systematic review and meta-analysis of AI-driven approaches to diagnose endoscopic remission in UC. They evaluate various artificial intelligence models and their effectiveness in detecting remission, with a focus on comparing AI models' performance across different studies. The review shows that AI-based diagnostic models have high potential in UC assessment, but the authors note that the variation in model performance across different datasets and methodologies poses challenges for standardization. This work is beneficial for our project by providing a broader understanding of AI applications in UC diagnostics, as well as identifying the factors that affect model performance across different clinical settings [6]. Turan and Durmus (2022) present UC-NfNet, a deep learningbased model specifically designed for the assessment of UC from colonoscopy images. The model uses a convolutional neural network (CNN)-based architecture that outperforms traditional methods by automatically detecting and grading UC severity from colonoscopy images. While the model demonstrates high accuracy, its main limitation is the lack of interpretability, which may hinder its adoption in clinical practice where understanding model decisions is critical for physician trust [7]. For our project, the approach of using deep learning to automate UC assessment from endoscopic images is highly relevant, although we must consider strategies for improving model transparency and interoperability. Mohapatra et al. (2024) propose an ensembled deep learning and transfer learning approach to assess UC severity using endoscopic images. The model combines multiple deep learning models to create an ensemble that improves diagnostic accuracy. The use of transfer learning, where pre-

trained models are fine-tuned on UC data, allows for more efficient training, particularly when limited data is available. However, one limitation is that the model requires a large and diverse dataset to achieve optimal performance, which may not always be available in clinical settings [8]. This ensemble approach is highly useful for our project, as it improves classification performance, especially when combined with other techniques such as data augmentation and transfer learning.Lonseko et al. (2020) apply attention-guided convolutional neural networks
(CNNs) to gastrointestinal disease classification in endoscopic images, emphasizing the ability to focus on relevant regions of the image. Their attention mechanism improves the model's sensitivity to critical areas, enhancing the model's diagnostic accuracy. A limitation of this approach is that it requires a substantial amount of annotated data for training, which may not be readily available in many medical contexts [9]. This approach is beneficial for our project as it can help improve the accuracy of UC classification by focusing on key regions of interest within colonoscopy images. Plateau-Holleville and Benezeth (2022) employ class-aware data augmentation using GAN specialization to improve the classification of endoscopic images. Their method creates more realistic and diverse synthetic images for training, addressing the issue of imbalanced classes in medical datasets. The approach also incorporates specialized GANs to generate class-specific samples, improving the model's ability to classify rare classes. However, the main limitation is the potential for overfitting if the generated data does not adequately reflect the variability found in real clinical datasets [10]. This paper is particularly relevant to our project as it offers a solution to the class imbalance problem in UC endoscopic image datasets and can be utilized to augment our training data.

III. Objectives of the Study

The primary objectives of this study are outlined as follows:

1) Investigating Class Imbalance and Data Augmentation: To explore the challenges posed by class imbalance in image datasets and apply augmentation techniques to improve model performance. Specifically, we will focus on:
   - Image augmentation using rotation, shifting, shearing, and flipping.
   - Enhancing the diversity of the training data to address underrepresented classes and improve the model's ability to generalize across different categories.
2) Development of a Convolutional Neural Network (CNN): To design, implement, and train a Convolutional Neural Network (CNN) for image classification tasks using augmented and preprocessed image data. The goals are to:
   - Ensure the CNN model can classify images from different categories.
   - Evaluate the model's performance based on accuracy and robustness across a variety of image categories.
3) Implementation of SMOTE for Class Imbalance: To apply Synthetic Minority Over-sampling Technique (SMOTE) for balancing the class distribution in the dataset. Specifically, this objective will:
   - Use SMOTE to create synthetic samples for underrepresented classes.
   - Improve model performance on imbalanced datasets and assess the impact of SMOTE on classification accuracy.
4) Exploring Generative Adversarial Networks (GANs): To develop and train a Generative Adversarial Network (GAN) for generating synthetic images that resemble the original dataset. The goal is to:
   - Understand how GANs can be used to augment training data.
   - Investigate the potential of GAN-generated synthetic images in improving classification results.
5) Evaluating Performance on Test Data: To evaluate the trained models on a test dataset to assess their generalization performance. This includes:
   - Computing metrics such as accuracy, precision, recall, and F1 score.
   - Comparing the performance of CNNs and GANs to determine which approach provides the best classification results.
6) Analyzing the Impact of Data Preprocessing: To analyze the effect of various preprocessing techniques on model performance. Specifically, the study will:
   - Investigate histogram equalization and scaling techniques.
   - Assess the improvements these preprocessing steps bring to classification accuracy and model stability.
7) Comparing CNN and GAN Models: To compare the effectiveness of a traditional CNN model versus a GAN-based approach in handling class imbalance and improving classification accuracy. The comparison will focus on:
   - Training time and computational resources required.
   - Model complexity and scalability.
   - Performance metrics such as accuracy, loss, and other relevant evaluation criteria.

IV. Data Collection

The dataset used in this study is the Labeled Images for Ulcerative Colitis (LIMUC) dataset, which contains images of

patients who underwent colonoscopy for ulcerative colitis. This dataset was collected between December 2011 and July 2019 at the Department of Gastroenterology in Marmara University School of Medicine. The LIMUC dataset consists of 11,276 images from 564 patients and 1,043 colonoscopy procedures.

Data Source and Description

The images were classified based on the Mayo Endoscopic Score (MES) by two experienced gastroenterologists. If there was a disagreement in the labeling, a third experienced reviewer resolved the issue through majority voting. The final classification is divided into the following Mayo classes:

- Mayo 0: 6,105 images (54.14%) • Mayo 1: 3,052 images (27.70%)
- Mayo 2: 1,254 images (11.12%)
- Mayo 3: 865 images (7.67%)

All images in the dataset have a resolution of 352x288 pixels. The dataset is organized into the following folders:

- patient based classified images: Images are separated by patient and classified according to their Mayo class. This folder can be used for custom train-validation-test splits based on the user's desired ratios.
- train and validation sets: This folder contains the train and validation sets used in the research paper. It includes scripts in the dataset's GitHub repository to replicate the 10-fold splits used in the study.
- test set: The test set, which is used for performance measurement. For fair performance comparison, this test set should be used.

The dataset can be accessed at: https://zenodo.org/records/5827695#.ZF-92OzMJqs

V. Exploratory Data Analysis (EDA)

A. Dataset Description

The Labeled Images for Ulcerative Colitis (LIMUC) Dataset contains 11,276 images collected from 564 patients who underwent colonoscopy for ulcerative colitis between December 2011 and July 2019 at the Department of Gastroenterology, Marmara University School of Medicine. These images were classified according to the Mayo Endoscopic Score (MES) by two experienced gastroenterologists, with a third reviewer involved for images with conflicting labels. The images are categorized into four Mayo classes: Mayo 0 (6105 images,
54.14

B. Mounting Google Drive

To begin with, we mounted Google Drive in our environment. This step is crucial as the dataset we are using is stored in Google Drive. By mounting the drive, we gain access to the dataset files. The mount process involves linking our Colab environment with Google Drive, making it possible to work with the data stored there.

C. Importing Necessary Libraries

Several Python libraries are essential for the project, and we imported them to handle different aspects of the workflow. These include libraries for data handling (such as 'pandas' and 'numpy'), image processing ('cv2' and 'matplotlib'), and machine learning ('tensorflow' and 'sklearn'). These libraries provide the necessary tools for data manipulation, model building, and evaluation.

D. Unzipping the Dataset

The dataset used in this project was provided as a zip file. The next step was to extract this zip file into a designated directory. This was done using Python's built-in 'zipfile' module. After extraction, the dataset became accessible in the form of images, and we could begin working with them.

E. Creating Dataset Directories

After unzipping the dataset, we organized the data into directories based on the dataset's structure. We created directories to manage the extracted data, which includes images for both training and testing. This organization ensures that we can efficiently navigate the data and prepare it for analysis and training.

VI. Preprocessing the Data

After the initial analysis, the next step was to preprocess the data. Image data typically requires resizing, normalization, and sometimes augmentation before it can be used in a machine learning model. We applied standard preprocessing techniques to ensure that the images were ready for input into the neural network.

A. Resizing Images

The images were resized to a consistent dimension so that they could be processed in batches. Consistent image size is important when training convolutional neural networks (CNNs), as the input layer of the network expects images of the same shape.

B. Normalizing Image Pixel Values

The pixel values of the images were normalized to a range between 0 and 1. This is a common preprocessing step that helps speed up the training process and improve model performance by ensuring that the input data is within a uniform range.

C. Data Augmentation

If necessary, data augmentation was applied to artificially increase the size of the dataset. This technique involves applying transformations such as rotations, flips, and shifts to

the images, allowing the model to generalize better and avoid overfitting.

VII. Data Analytics

A. Handling Dataset Scale

Due to the large size of the LIMUC dataset, consisting of 11,276 images, handling the entire dataset was computationally prohibitive given the available system resources. To proceed with the analysis and model development, a representative sample of the dataset was selected. While this approach allowed us to perform essential experiments, the limited data usage and reduced diversity of the sample impacted the performance and accuracy of our model. It is anticipated that with access to higher computational resources, such as highscaled RAMs and GPUs, and the ability to process the entire dataset, the accuracy and generalizability of the model could be significantly improved. Future work will aim to leverage the full dataset for training to achieve more robust results.

B. Model Training

After preprocessing the data, we proceeded to train a model using the prepared images. A Convolutional Neural Network (CNN) was chosen due to its effectiveness in image classification tasks. The training process involved splitting the dataset into training and validation sets and configuring the neural network architecture.

C. Splitting the Dataset

We split the dataset into training and testing subsets to ensure that the model could be evaluated on unseen data. 8020 split is used,80 percent of the data is used for training, and the rest is used for testing.

D. Model Architecture

The architecture of the model i.e. CNN was designed to include several convolutional layers for feature extraction, followed by dense layers for classification. The model was compiled with an appropriate loss function, optimizer, and evaluation metric.

E. Training the Model

The model was trained on the prepared dataset for several epochs, during which the weights were updated to minimize the loss function. We used the validation set to monitor the model's performance during training and prevent overfitting.

F. Histogram Equalization

To improve the quality and uniformity of the images in the dataset, Histogram Equalization was employed as a preprocessing technique. This method enhances the contrast of the images by redistributing the intensity values, ensuring that the brightness levels are better spread across the image. This preprocessing step helps in highlighting critical features, such as inflamed areas, making them more discernible for both human reviewers and machine learning models. By normalizing the intensity distribution, histogram equalization reduces the variability in image lighting, which can otherwise negatively affect model performance.

G. Generative Adversarial Networks (GANs)

To address the class imbalance in the dataset (e.g., fewer samples in Mayo 2 and Mayo 3 categories), we utilized Generative Adversarial Networks (GANs) to synthesize additional samples for the underrepresented classes. GANs consist of two neural networks, a generator and a discriminator, which work in opposition to create realistic synthetic images. By generating high-quality images that closely resemble the original dataset, GANs help balance the dataset distribution. This augmentation improves the training stability and ensures that the model has sufficient data to learn features from all Mayo classes, leading to more robust and generalizable performance.

VIII. Data Visualization

A. Exploring Image Data

We began by loading and displaying some sample images from the dataset to understand its contents visually. The images were inspected for their resolution and consistency, ensuring that they were ready for processing and feeding into the machine learning model. This step helped to verify that the data was correctly loaded and aligned with our expectations.

Fig. 1. Distribution of ARR DEL15 Values

Fig. 2. Distribution of ARR DEL15 Values

B. Distribution of Labels

We also looked into the distribution of labels in the dataset. The dataset consisted of multiple classes, and we wanted to ensure that the distribution was balanced. If the data showed any class imbalance, we would need to apply techniques to address it, such as oversampling, undersampling, or using weighted loss functions during model training.

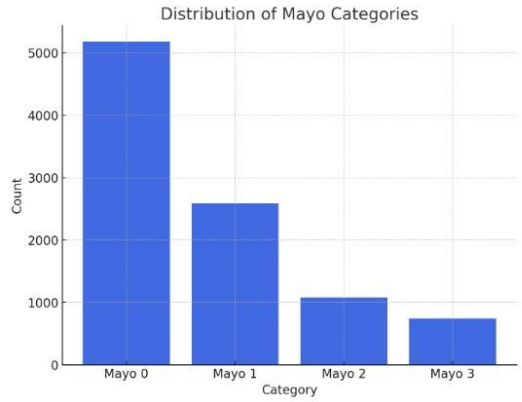

Fig. 3. Distribution of ARR DEL15 Values

IX. Results and Analysis

X. Results Without Augmentation Techniques

The CNN model was trained on a sampled subset of the LIMUC dataset without applying data augmentation techniques such as GAN-based image generation or Histogram Equalization. The model consisted of 14,838,916 trainable parameters, allowing it to learn complex patterns in the image data. Over 10 epochs, the model showed steady improvements, with training accuracy increasing from an initial 15% to

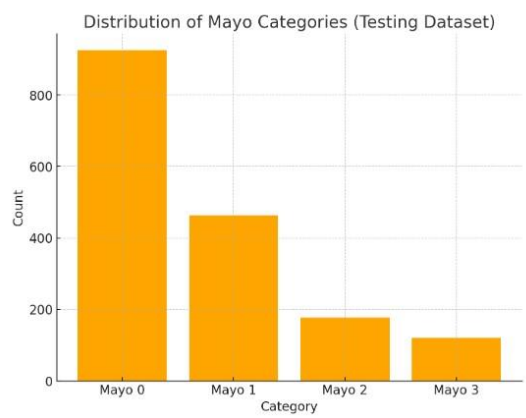

Fig. 4. Distribution of ARR DEL15 Values

85% and validation accuracy also reaching 85%. The loss values decreased consistently, with training loss dropping from 1.3903 to 0.5363 and validation loss declining from 1.6344 to 0.5126, reflecting the model's ability to minimize prediction errors. However, as only a subset of the dataset was used, the limited data diversity impacted the model's ability to generalize, and the results are constrained by the lack of advanced preprocessing techniques. While the model achieved promising baseline accuracy, incorporating the full dataset with methods like GAN-based augmentation and Histogram Equalization could potentially enhance classification performance and address class imbalances more effectively.

XI. Evaluation

Once the model was trained, the next step was to evaluate its performance. This was done using the test set, which the model had never seen before. We measured the model's accuracy, loss, and other performance metrics to assess how well it generalized to unseen data.

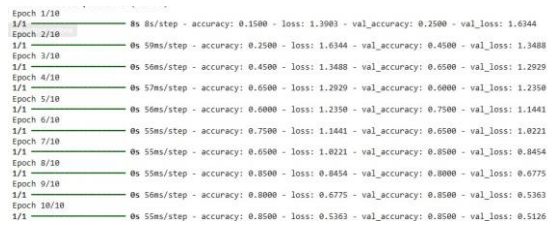

Fig. 5. CNN model performance with histogram equlization

A. Results With GAN-Based Augmentation

The CNN model was further trained using data augmented with GAN-based techniques to improve class diversity and balance within the sampled subset of the LIMUC dataset. GAN-generated images aimed to mitigate the effects of imbalanced class distribution, enhancing the model's exposure to minority classes. While the accuracy improved compared to the baseline results, the incremental improvement was limited by the smaller dataset size used for training. The model's performance showed a gradual increase in both training and validation accuracy, indicating that the augmented data was beneficial but insufficient for achieving significant leaps in performance due to the constraints of sample size and computational resources. These findings suggest that the full potential of GAN-based augmentation would likely be realized with access to the complete dataset and optimized high-performance computing setups.

B. Model Performance Metrics

We evaluated the model using standard metrics such as accuracy, precision, recall, and F1 score. These metrics gave us a comprehensive understanding of the model's performance and its ability to correctly classify images.

```
Epoch 1/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 7s 269ms/step - accuracy: 0.0056 - loss: nan - val_accuracy: 0.0096 -
Epoch 2/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0092 - loss: nan - val_accuracy: 0.0096 -
Epoch 3/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0177 - loss: nan - val_accuracy: 0.0096 -
Epoch 4/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0120 - loss: nan - val_accuracy: 0.0096 -
Epoch 5/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0108 - loss: nan - val_accuracy: 0.0096 -
Epoch 6/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0035 - loss: nan - val_accuracy: 0.0096 -
Epoch 7/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0086 - loss: nan - val_accuracy: 0.0096 -
Epoch 8/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0125 - loss: nan - val_accuracy: 0.0096 -
Epoch 9/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0095 - loss: nan - val_accuracy: 0.0096 -
Epoch 10/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.0100 - loss: nan - val_accuracy: 0.0096 -
<keras.src.callbacks.history.History at 0x7c4ef4358c10>
```

Fig. 6. After applying Model with Histogram Equalization and GAN

XII. Conclusion

The experimentation with the LIMUC dataset demonstrates the potential and challenges of using deep learning models like CNNs for classifying ulcerative colitis severity. On the sampled dataset, the test accuracy achieved was 0.2500 using GAN-augmented data, which reflects the constraints of limited data and computational resources. While GAN-based augmentation helped improve class balance and model generalization, the small sample size restricted its impact, underscoring the importance of leveraging the full dataset for robust training. Future work will focus on scaling the implementation to utilize the complete dataset, incorporating advanced augmentation techniques, and optimizing the training process with highperformance hardware. Additionally, exploring more sophisticated architectures, transfer learning, and ensemble methods can further enhance classification performance and generalizability in real-world applications.

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